

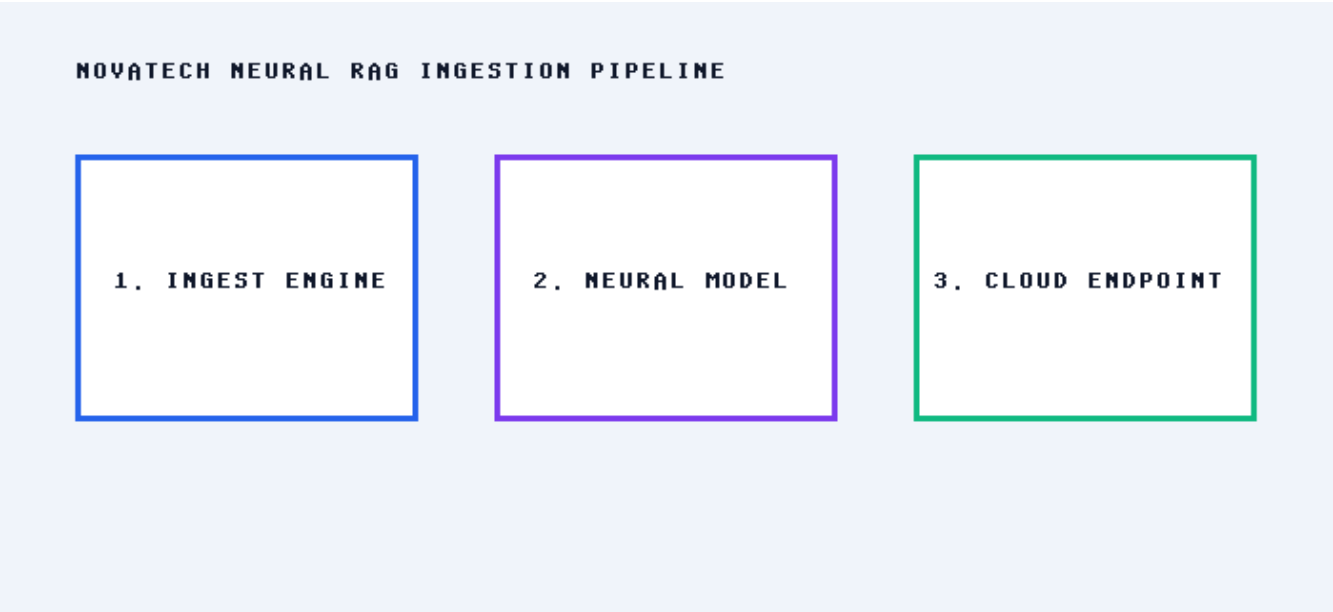
NovaTech AI & ML Technical Guide

Architecting Enterprise RAG & Neural Networks

1. Introduction to Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is an architectural framework that enhances Large Language Model (LLM) responses by dynamically referencing authoritative external knowledge bases. At NovaTech Solutions, our RAG framework incorporates multi-modal document ingestion, vector databases (such as Pinecone, Qdrant, and Milvus), and semantic chunking algorithms.

2. End-to-End Ingestion Pipeline Diagram



3. Key Technical Specifications

Chunking Strategy: Recursive Character Splitter (512 tokens with 50-token overlap).
Embedding Model: text-embedding-3-large (1536 dimensions).
Vector Search Index: HNSW (Hierarchical Navigable Small World) cosine distance.
Reranking Algorithm: Cohere Rerank v3 for top-k contextual relevance scoring.